

TheAnalyticsTeam

Sprocket Central Pty Ltd

Data analytics approach

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Agenda

1. Introduction
2. Data Exploration
3. Model Development
4. Interpretation

Introduction

Problem Outline

- ❖ Sprocket Central Pty Ltd is a long-time customer of KPMG that specialises in selling high-quality bikes and affordable cycling accessories to cyclists.
- ❖ The Company is targeting 1000 new Customers to drive the most value for the organisation.
- ❖ Sprocket Central Pty Ltd marketing team is looking to boost business by analysing their existing customer dataset to determine customer trends and behaviour.

Approach for Customer Analysis

- ❖ Age Distributions
- ❖ Number of bike purchased in Previous 3 years
- ❖ Job industry segmentation
- ❖ Wealth Segmentation
- ❖ Cars Owner Segmentation in each state
- ❖ Customer Profile

Data Exploration

DATA QUALITY ISSUE HIGHLIGHTS IN TRANSACTION, CUSTOMER DEMOGRAPHICS AND CUSTOMER ADDRESS DATA SETS

Sheet name	Column Name	Issues	Data Quality Dimensions	Details
Transaction	Online order	Null Value	Completeness	Blanks
	Brands	Null Value	Completeness	Blanks
	Product First Sold Date	Meta-data values	Relevancy	Convert (int) to Date Format
	Cancelled Status	Filtered out	Relevancy	Order status
	Profit Column missing	Profit/Revenue missing	Accuracy	calculated Revenue as was missing
	List price & Product Sold	Format	Validity	Data Formatting
NewCustomerList	Last name	Null Value	Completeness	Blanks
	Gender	U as Gender	Consistency	Discarded value in gender as no gender equals "U"
	Job title	Null Value	Completeness	Blanks in job title cell
	Job Industry	Null Value	Completeness	N/A values in cell
Customer Address	States		Consistency	States name are not Correct
CustomerDemographics	Last name	Null Value	Completeness	Blanks
	Gender	U as Gender	Consistency	Spelling error Gender misspelled as "U"
	DOB	Null Value	Completeness	Missing values
	Job title	Null Value	Completeness	Blanks in job title cell
	Job Industry Category	Null Value	Completeness	N/A values in cell
	deceased indicator		Consistency	Removed "Y" from column as we require data from people who are alive
	default		Relevancy	doesn't Relate with data

Data Exploration

Bike Related Purchase Based on Age And Gender of customers

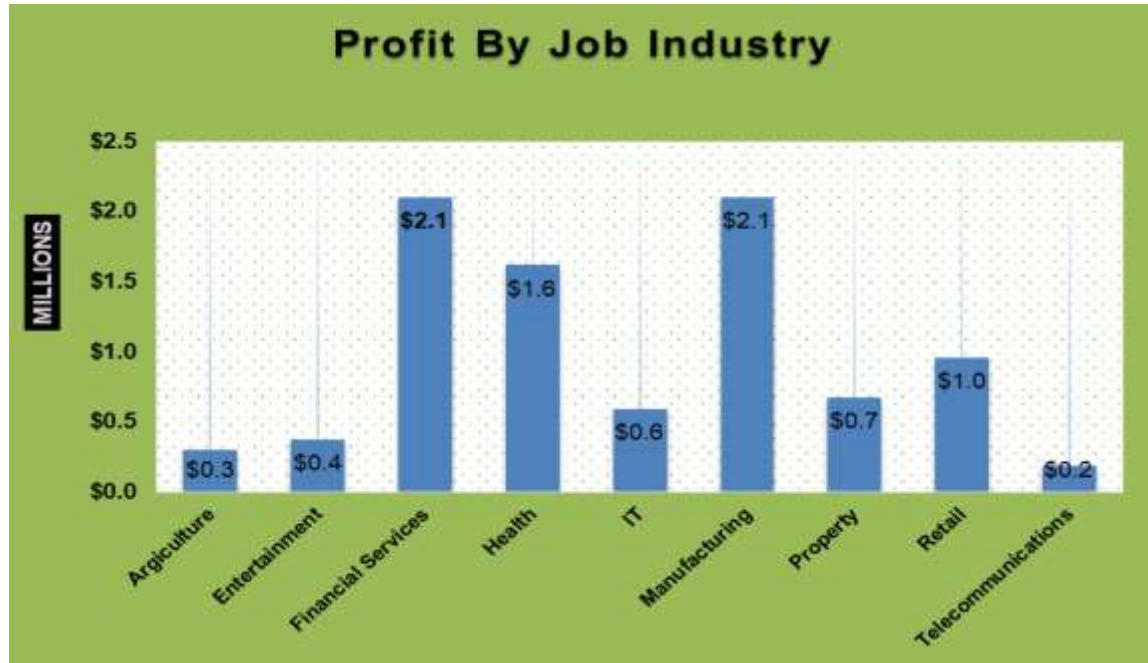
- ❖ Highest Bike buying customers are between 40-49 age group.
- ❖ In the new purchase customer Graph people age between 40-49 and 50-59 are the highest bike purchasers.
- ❖ In the Bike purchase by gender Females are the highest bike purchasers.
- ❖ The visualisation shows high purchase numbers between middle aged group and Females which declares them as Potential Customers.



Data Exploration

Top Performing Job industries sectors Related to Max Profit and Bike Purchasing

- ❖ Top 3 industry contributing to highest profit are: **Financial Services, Health, Manufacturing** .
- ❖ The job industries sectors are within city commute, which is reason of high bike purchase.
- ❖ Agriculture, Entertainment, IT, Property, Retail and Telecommunications have returned less than \$1Millon in profits.

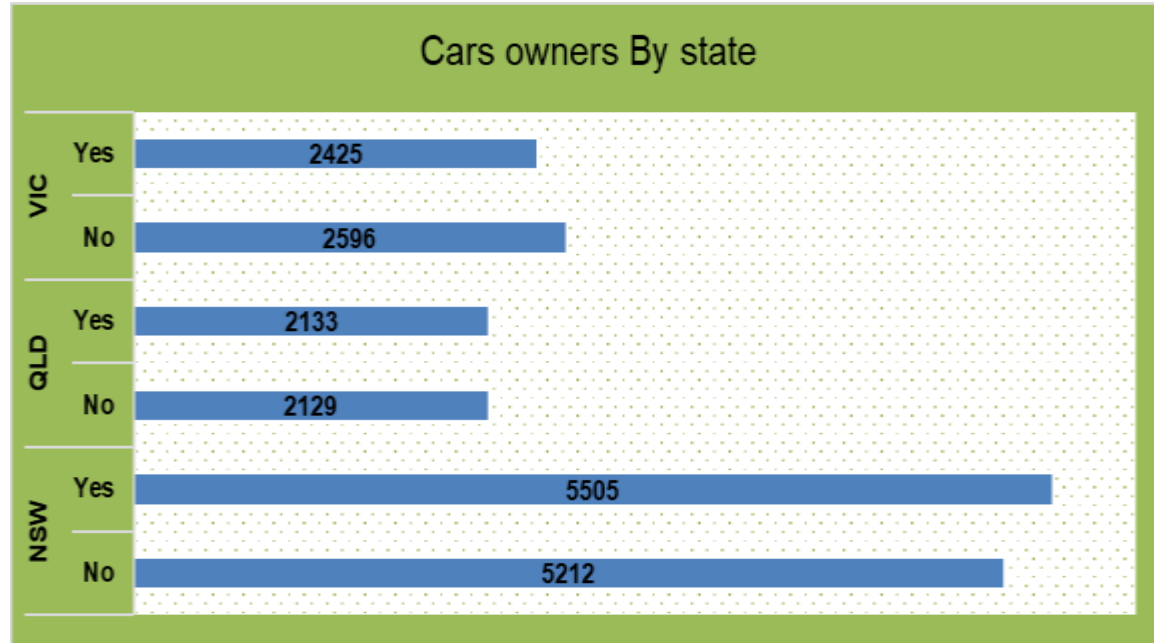


Data Exploration

No of Cars owned in each State

❖ States namely **NSW**, **QLD**, **VIC** could be premium markets which have potential.

❖ NSW has almost the equal no of bike Purchasers who owns cars and who don't have no cars which indicates **high potential of market growth**.



Customer Classification – Targeting high Value Customers

High values customers who will be targeted from new list.

- Mostly high value customers are female as compared to males
- Working in Health, Financial, Manufacturing industry sector.
- Aged between 38-49.
- Who are based in NSW, VIC.

Interpretation

Summary Table of High Valued customers

Here are some high valued customers which come under Classification of Platinum Customers.

First Name	Age	gender	past_3_years_bike_related_purchase	job_industry_category	owns_car	State	Customer Profile	wealth_segment	RFM Total
Marcellus	27	Male	76	Manufacturing	No	VIC	Platinum	Mass Customer	444
Andonis	65	Male	21	Property	No	NSW	Platinum	Mass Customer	444
Paulie	45	Female	44	Financial Services	No	VIC	Platinum	Mass Customer	444
Karissa	25	Female	10	Property	No	NSW	Platinum	Mass Customer	444
Stavro	42	Male	19	IT	Yes	VIC	Platinum	Mass Customer	443
Tamarah	43	Female	43	Manufacturing	Yes	QLD	Platinum	Mass Customer	434
Carl	41	Male	51	Manufacturing	No	NSW	Platinum	Mass Customer	434
Obed	34	Male	12	IT	Yes	NSW	Platinum	Mass Customer	432
Cherin	33	Female	50	Manufacturing	No	NSW	Platinum	Mass Customer	422
Brittany	21	Female	98	Argiculture	Yes	NSW	Platinum	Mass Customer	422
Rolando	41	Male	62	Property	No	NSW	Platinum	Mass Customer	422

Thank You!

Appendix

CFM Analysis Customer Profile Segmentation

Row Labels	Count of RMF(Total)
Bronze	1014
Gold	875
Platinum	797
Silver	809

